

Question 1: String Functions

Write a Java Program to do the following:

- ➔ Prompt the user to enter their first name and age using Scanner.
- ➔ Reverse the first name using “StringBuilder” class and print the reversed name
- ➔ If age of the user is above or equal to 18, print “User is an adult”. Otherwise, “User is a kid”.

Sample Input:

Enter your name: Hari

Enter your age: 39

Sample Output:

Your reversed name is iraH.

You are an adult.

Question 2: String Functions

Write a Java Program to do the following:

- ➔ Prompt the user to enter an email ID
- ➔ Check whether the given email ID is valid

Sample Input I:

Enter your email ID: abc.def@xyz.com

Sample Output I:

Your email ID is valid

Sample Input II:

Enter your email ID: [abc.def@xyz](#)

Sample Output II:

Your email ID is invalid

Question 3: String Functions + Regular Expressions

Write a Java Program to do the following:

- ➔ Prompt the user to enter an User ID
- ➔ Check whether the given user ID is valid by following the conditions
 - At least 8 characters
 - Should contain at least one uppercase character
 - Should contain at least one digit

Sample Input I:

Enter your user ID: Haric2025

Sample Output I:

Your user ID is valid

Sample Input II:

Enter your user ID: Haric25

Sample Output II:
Your user ID is invalid

Question 4: ArrayList Functions

Write a Java Program to do the following:

- ➔ Create an ArrayList to store integers.
- ➔ Add 10 random integers (between 1 and 100) to the ArrayList.
- ➔ Sort the ArrayList in ascending order and display the sorted list.
- ➔ Find and display the maximum and minimum values in the ArrayList.
- ➔ Remove all even numbers from the ArrayList and display the remaining elements.
- ➔ Check if a specific integer (input by the user) is present in the ArrayList and display the result.
- ➔ Clear the ArrayList and display its size after clearing.

Sample Output:

Original ArrayList: [23, 45, 12, 67, 89, 34, 56, 78, 90, 11]

Sorted ArrayList: [11, 12, 23, 34, 45, 56, 67, 78, 89, 90]

Maximum value: 90

Minimum value: 11

ArrayList after removing even numbers: [23, 45, 67, 89, 11]

Enter an integer to check: 45

Is 45 present in the ArrayList? true

ArrayList size after clearing: 0

Question 5: OOPs

Write a Java program to create a class called "Employee" with a name, job title, and salary attributes, and methods to update the salary for a given %.

Sample Output:

Employee Details:

Name: Sachin Tendulkar

Job Title: Manager

Salary: 15000.0

Name: Saurav Ganguly

Job Title: Software Engineer

Salary: 30000.0

After raising salary:

8% for ' Sachin Tendulkar':

Name: Sachin Tendulkar

Job Title: Manager

Salary: 16200.0

12% for 'Saurav Ganguly':

Name: Saurav Ganguly

Job Title: Software Engineer

Salary: 33600.0

Question 6: OOPs – II

Write a Java program to create a class called "Shape" with abstract methods for calculating area and perimeter, and subclasses for "Rectangle", "Circle", and "Triangle".

Question 7: Enum

Write a Java program to do the following:

- ➔ Define an Enum called Day with the days of the week (MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, SUNDAY).
- ➔ Create a method that takes a Day enum as a parameter and returns a message indicating whether it's a weekday or weekend.
- ➔ Iterate over all values of the Day enum and display each day along with whether it's a weekday or weekend.
- ➔ Prompt the user to input a day (as a string) and display the corresponding enum value and message.

Sample Output:

MONDAY is a weekday.

TUESDAY is a weekday.

WEDNESDAY is a weekday.

THURSDAY is a weekday.

FRIDAY is a weekday.

SATURDAY is a weekend.

SUNDAY is a weekend.

Enter a day: Friday

FRIDAY is a weekday.

Question 8: Custom Exceptions

Write a Java program to do the following:

- ➔ Define a custom exception called InvalidAgeException that extends Exception.

- ➔ Create a method called `checkAge` that takes an integer `age` as a parameter and throws `InvalidAgeException` if the age is less than 18.
- ➔ Prompt the user to input an age and use the `checkAge` method to validate the age.
- ➔ Catch the custom exception and display an appropriate message if the age is invalid.
- ➔ Display a success message if the age is valid.

Sample Output:

Enter your age: 16

`InvalidAgeException`: Age must be 18 or older.

Enter your age: 20

Age is valid.

Question 9: `Stream()` and `map()`

Write a Java program to do the following:

- ➔ Create a list of integers containing 15 random integers between 1 and 50.
- ➔ Display the original list of integers.
- ➔ Use the `map` method to create a new stream that squares each integer.
- ➔ Collect the mapped stream into a new list and display the list of squared integers.

Sample Output:

Original list: [3, 12, 25, 7, 18, 30, 45, 9, 21, 14, 5, 27, 33, 11, 19]

List of squared integers: [9, 144, 625, 49, 324, 900, 2025, 81, 441, 196, 25, 729, 1089, 121, 361]

Question 10: File Access

- ➔ **Create an Excel file** with the following data:

Name	Age	City	
Alice	30	New York	
Bob	25	Los Angeles	
Charlie	35	Chicago	
David	28	Houston	
Eve	22	Phoenix	

- ➔ Define a class called `Person` with the following attributes:

```
String name
int age
String city
```

- ➔ Create a method in the `Person` class to display the person's details.

➔ Write a Java program to read the data from the Excel file, create Person objects for each row, and store them in a list.

1. Display the details of each Person object using the method defined in the Person class.
2. Use Apache POI library to handle the Excel file operations.

Sample Output:

Name: Alice, Age: 30, City: New York

Name: Bob, Age: 25, City: Los Angeles

Name: Charlie, Age: 35, City: Chicago

Name: David, Age: 28, City: Houston

Name: Eve, Age: 22, City: Phoenix